

DAY 5

Financial Modeling for Mergers and Acquisitions (M&A) and Investment Decisions

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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This chapter builds working competence in Financial Modeling for Mergers and Acquisitions (M&A) and Investment Decisions: the concepts, the standard procedure, and the judgment to apply it under exam conditions.

Concepts and vocabulary

Financial modeling in the M&A and investment context is the practice of building structured, quantitative representations of a company's financial reality—past, present, and projected—to support high-stakes decisions such as acquisitions, divestitures, capital raises, and strategic investments. Before diving into techniques, every practitioner must command the vocabulary precisely, because imprecise language leads to imprecise models, and imprecise models lead to costly errors.

Core Definitions

Financial model: A spreadsheet-based (or programmatic) framework that links assumptions to outputs through logical, auditable formulas. In M&A, the model typically integrates three interconnected financial statements—income statement, balance sheet, and cash flow statement—and extends them into a forecast horizon.

Target company: The firm being acquired or evaluated for investment. The acquirer (or buyer) is the entity making the offer.

Synergies: Incremental value created when two companies combine that neither could achieve independently. Revenue synergies arise from cross-selling or expanded market reach; cost synergies arise from eliminating redundant expenses.

Enterprise Value (EV): The total value of a firm's operations, independent of capital structure.

$$EV = \text{Equity Value} + \text{Net Debt}$$

where Net Debt = Total Debt " Cash and Cash Equivalents.

Equity Value (Market Capitalization): The value attributable to shareholders alone.

$$\text{Equity Value} = EV - \text{Net Debt}$$

EBITDA: Earnings Before Interest, Taxes, Depreciation, and Amortization—a widely used proxy for operating cash generation.

EV/EBITDA multiple: A valuation benchmark comparing enterprise value to operating earnings.

$$EV/EBITDA = \text{Enterprise Value} \div EBITDA$$

Accretion/Dilution: A deal is accretive if it increases the acquirer's earnings per share (EPS) post-close; dilutive if it decreases EPS. This is one of the first tests applied to any proposed transaction.

Discount rate (WACC): The Weighted Average Cost of Capital, used to convert future cash flows into present value.

$$WACC = (E/V) \times R' + (D/V) \times R_H \times (1 - t)$$

where E = equity, D = debt, V = total capital, R^E = cost of equity, R^D = pre-tax cost of debt, and t = tax rate.

Where Financial Modeling Fits in Practice

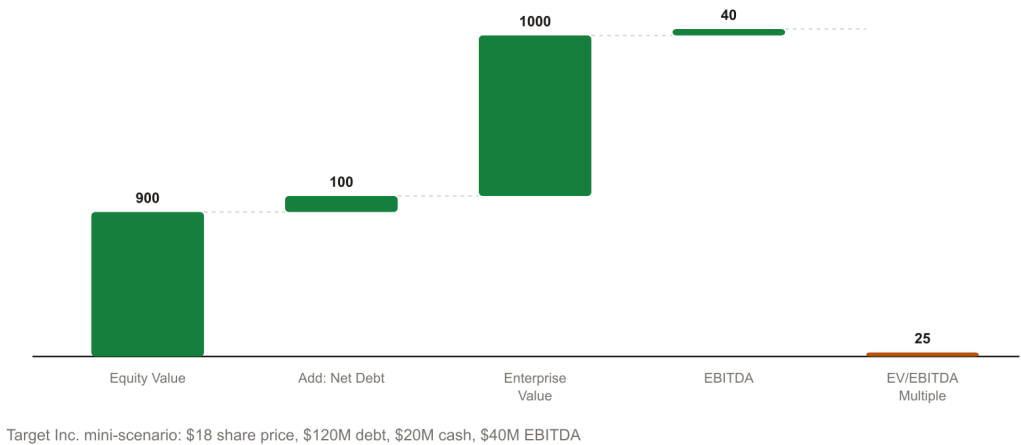
In corporate finance and professional accounting, financial modeling sits at the intersection of valuation, decision-making, and communication. An analyst building an M&A model is simultaneously a forecaster, a storyteller, and a risk assessor. Models are used to:

- Determine a fair offer price for a target
- Stress-test deal assumptions under bear, base, and bull scenarios
- Satisfy due diligence requirements for boards, lenders, and regulators
- Illustrate capital requirements to investors or financing partners

Practitioners carrying titles such as financial analyst, associate, or vice president in investment banking, corporate development, or private equity are the primary builders and consumers of these models.

Standard Notation

Enterprise Value Build-Up



Target Inc. mini-scenario: \$18 share price, \$120M debt, \$20M cash, \$40M EBITDA

| Symbol | Meaning |

|-----|-----|

| EV | Enterprise Value |

| Mkt Cap | Market Capitalization (Equity Value) |

| ND | Net Debt |

| EBITDA | Earnings Before Interest, Taxes, Depreciation & Amortization |

| WACC | Weighted Average Cost of Capital |

| FCF | Free Cash Flow |

| PV | Present Value |

| t | Corporate tax rate |

Worked Example: Terminology Applied to a Mini Scenario

Scenario: Acquirer Corp is evaluating the purchase of Target Inc., a manufacturing firm.

Given information:

- Target's share price: \$18.00; shares outstanding: 50 million
- Target's total debt: \$120 million; cash on hand: \$20 million
- Target's trailing EBITDA: \$40 million
- Acquirer's WACC: 9%

Step 1 — Calculate Equity Value (Market Capitalization):

$$\text{Equity Value} = \$18.00 \times 50,000,000 = \$900,000,000 = \$900\text{M}$$

Step 2 — Calculate Net Debt:

$$\text{Net Debt} = \$120\text{M} - \$20\text{M} = \$100\text{M}$$

Step 3 — Calculate Enterprise Value:

$$\text{EV} = \$900\text{M} + \$100\text{M} = \$1,000\text{M}$$

Step 4 — Calculate the EV/EBITDA multiple:

$$\text{EV/EBITDA} = \$1,000\text{M} \div \$40\text{M} = 25.0\times$$

Step 5 — Interpret: A 25× EV/EBITDA multiple tells Acquirer Corp how the market prices each dollar of Target's operating earnings. The analyst would now compare this to industry peers (e.g., if comparable companies trade at 18–22×, Target appears richly priced) and run a discounted cash flow model using the 9% WACC to triangulate intrinsic value.

Common mistake: Analysts frequently confuse Equity Value with Enterprise Value, especially when quoting deal size. A \$1 billion "deal" may refer to the equity check written, but the true economic cost includes assumed debt. Always clarify which metric is being cited before building or presenting any model.

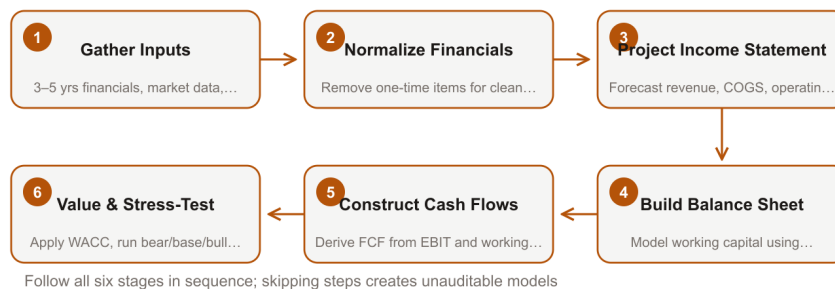
Mastering these definitions is not merely academic—every formula in an M&A model traces back to these building blocks. A solid vocabulary ensures that when assumptions change, you know exactly which output is affected and why.

The standard procedure

The standard procedure for M&A and investment financial modeling follows a disciplined, repeatable sequence. Mastering each stage—inputs, steps, outputs, and checks—gives you a model that is both defensible in a boardroom and auditable by a third party.

Inputs

M&A Modeling Procedure



Follow all six stages in sequence; skipping steps creates unauditable models

Every model begins with raw material gathered from three sources: historical financials (at least three to five years of income statements, balance sheets, and cash flow statements), market and industry data (growth rates, comparable transaction multiples, discount rates), and deal-specific assumptions (purchase price, financing structure, synergy estimates, and integration costs). Label every assumption explicitly and store them on a dedicated inputs tab. This single habit prevents the most common source of modeling errors: buried hard-coded numbers scattered across worksheets.

Steps

Follow this sequence without skipping stages:

- Normalize historical financials. Strip out one-time items—litigation settlements, asset write-downs, restructuring charges—so the baseline reflects true operating performance.
- Project the income statement. Drive revenue from a bottom-up or top-down growth assumption, then forecast COGS and operating expenses as percentages of revenue or via unit-cost logic.
- Build the balance sheet and working capital schedule. Forecast receivables, inventory, and payables using days-outstanding metrics derived from history.
- Construct the cash flow statement. Derive free cash flow (FCF) from EBIT, apply taxes, add back depreciation and amortization, and subtract capital expenditures and changes in working capital.
- Model the deal structure. Layer in the acquisition price, goodwill, new debt, and any equity issuance. Adjust the pro-forma balance sheet accordingly.

- Incorporate synergies and integration costs. Phase synergies in over a realistic ramp period (commonly years one through three) and front-load integration costs in year one.
- Run the DCF valuation. Discount projected FCFs at the weighted average cost of capital (WACC) and add a terminal value.
- Perform accretion/dilution analysis. Compare pro-forma earnings per share (EPS) to standalone EPS to determine whether the deal is accretive or dilutive to the acquirer.

Worked Example

Scenario: AcquireCo is purchasing TargetCo for \$500 million. TargetCo's current EBIT is \$60 million, with a 25% tax rate, \$10 million in D&A, \$8 million in capex, and a \$4 million increase in working capital. WACC is 9%. FCF is assumed to grow at 3% in perpetuity after a five-year explicit forecast.

Step 1 – Calculate Year 1 FCF:

$$\text{FCF} = \text{EBIT} \times (1 - \text{Tax Rate}) + \text{D\&A} - \text{Capex} - \Delta \text{WC}$$

$$\text{FCF} = \$60\text{M} \times (1 - 0.25) + \$10\text{M} - \$8\text{M} - \$4\text{M} = \$45\text{M} - \$8\text{M} - \$4\text{M} = \$33\text{M}$$

Step 2 – Project FCFs for years 2–5 at a modest 5% annual growth (reflecting synergies phasing in):

Year	FCF
1	\$43.0M
2	\$45.2M
3	\$47.4M
4	\$49.8M
5	\$52.3M

Step 3 – Calculate Terminal Value (Gordon Growth Model):

$$\text{TV} = \frac{\text{FCF}_{\text{Year 5}} \times (1 + g)}{\text{WACC} - g} = \frac{\$52.3\text{M} \times 1.03}{0.09 - 0.03} = \frac{\$53.9\text{M}}{0.06} = \$898\text{M}$$

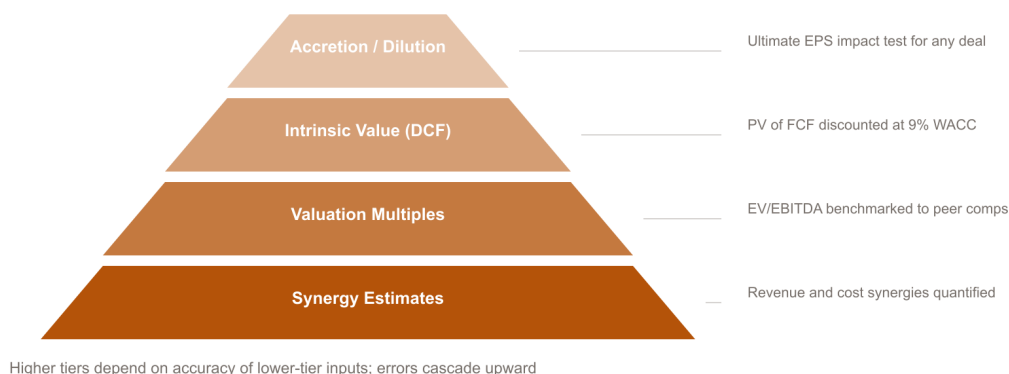
Step 4 – Discount all cash flows at 9%:

Sum of PV of years 1–5 = \$185M; PV of terminal value = \$584M; Enterprise Value = \$769M

Step 5 – Compare to purchase price: \$769M > \$500M, indicating the deal creates value at these assumptions.

Outputs

M&A Model Output Hierarchy



Higher tiers depend on accuracy of lower-tier inputs; errors cascade upward

The model should produce: a pro-forma three-statement model, a DCF valuation summary, an accretion/dilution table, and a sensitivity matrix varying WACC and growth rate across a realistic range.

Checks

- Circular reference audit: Interest expense feeding into net income and then into debt repayment must be handled with an iteration switch.
- Balance sheet balance: Assets must equal liabilities plus equity in every projected period—no exceptions.
- Sanity check multiples: Implied EV/EBITDA should fall within the range of comparable transactions.
- Sensitivity stress test: If a 1% rise in WACC flips the deal from value-creating to value-destroying, flag that fragility explicitly.

Common mistake: Modelers frequently apply synergies at full run-rate from year one. Synergies almost always ramp gradually; front-loading them inflates the valuation and can cause a deal to appear far more attractive than it truly is. Always phase synergies across at least two to three years and document the rationale for each assumption.

Applications and pitfalls

Financial models sit at the heart of every M&A transaction and major capital investment decision. They translate strategic assumptions into numbers that boards, lenders, and investors can scrutinize—and challenge. Understanding where these models are typically deployed, where analysts go wrong, and how to verify results before presenting them is essential for anyone working toward a finance certification or a career in corporate development.

Typical Use Cases

In M&A, financial models serve three overlapping purposes: valuation, deal structuring, and synergy quantification. A discounted cash flow (DCF) model establishes an intrinsic value range for the target. A leveraged buyout (LBO) model tests whether a private-equity sponsor can achieve its return hurdle

given a specific debt structure. An accretion/dilution model answers the immediate question every public-company board asks: does this deal increase or decrease earnings per share in year one?

For investment decisions outside M&A—capital expenditure approvals, greenfield projects, joint ventures—the same DCF logic applies, but the focus shifts to net present value (NPV) and internal rate of return (IRR). The model must capture incremental cash flows only, exclude sunk costs, and reflect the project's own risk profile in the discount rate.

Common Mistakes

The most damaging errors are not arithmetic; they are structural. Analysts frequently:

- Mix nominal and real cash flows. Projecting revenues in today's dollars while using a nominal discount rate double-counts inflation.
- Ignore working-capital changes. A fast-growing target consumes cash in receivables and inventory; omitting this overstates free cash flow.
- Apply the acquirer's WACC to the target. The discount rate must reflect the target's risk, not the buyer's.
- Double-count synergies. Synergies already embedded in the purchase price should not reappear as incremental value in the integration model.
- Use a terminal-value growth rate above long-run GDP. This implies the company eventually becomes larger than the economy—a mathematical absurdity.

A Flawed Attempt, Corrected Step by Step

Scenario: An analyst values a manufacturing target with \$50 million in Year 1 free cash flow (FCF), projecting 4% annual growth for five years, then applying a terminal value. The acquirer's WACC is 8%; the target's WACC is 10%. The analyst uses 8% throughout and forgets to include a \$6 million annual increase in net working capital (NWC).

Step 1 — Identify the correct discount rate.

The target carries more operational risk. Use 10%, not 8%.

Step 2 — Rebuild FCF to include working-capital drag.

$$\text{Adjusted FCF} = \text{Reported FCF} - \Delta \text{NWC}$$

Year 1 adjusted FCF = \$50M - \$6M = \$44M

Step 3 — Project adjusted FCF for Years 1–5 at 4% growth.

| Year | Adjusted FCF (\$M) |

|-----|

| 1 | 44.00 |

| 2 | 45.76 |

| 3 | 47.59 |

| 4 | 49.49 |

| 5 | 51.47 |

Step 4 — Calculate terminal value using the Gordon Growth Model.

$$\text{Terminal Value} = \text{FCF}_{\text{Year 5}} \times (1 + g) / (\text{WACC} - g)$$

$$\text{TV} = 51.47 \times 1.03 / (0.10 - 0.03) = 53.01 / 0.07 = \$757.3\text{M}$$

Step 5 — Discount all cash flows at 10%.

$$\text{PV} = \text{FCF}_1/1.10 + \text{FCF}_2/1.10^2 + \text{FCF}_3/1.10^3 + \text{FCF}_4/1.10^4 + \text{FCF}_5/1.10^5 + \text{TV}/1.10^5$$

PV of Years 1–5 = \$44.0/1.10 + \$45.76/1.21 + \$47.59/1.331 + \$49.49/1.464 + \$51.47/1.611 = \$176.5M

PV of terminal value = \$757.3M / 1.611 = \$470.1M

Enterprise Value = \$646.6M

The original flawed model—using 8% and ignoring NWC—produced an enterprise value near \$820M, a 27% overstatement that could have led to a significant overpayment.

Sanity Checks

Before any model leaves your desk, run these tests:

- Comparable-company cross-check: Does your implied EV/EBITDA multiple fall within the peer range? A result of 18× when peers trade at 9–11× demands explanation.
- Terminal-value proportion: If TV exceeds 80% of total enterprise value, your near-term projections may be too conservative or your growth rate too aggressive.
- Circular-reference audit: Interest expense linked to a revolving credit facility can create loops that silently corrupt outputs.
- Stress test the WACC: A 1% rise in the discount rate should move value meaningfully; if it barely budges, check whether the terminal growth rate is too close to WACC.

Common mistake: Analysts routinely set the terminal growth rate equal to the projected revenue growth rate used in the explicit forecast period. These are different concepts. The terminal rate should reflect long-run sustainable growth—typically 2–3% for a mature business—not the optimistic near-term trajectory baked into the five-year plan.

Disciplined application of these checks transforms a financial model from a persuasive-looking spreadsheet into a defensible analytical tool that withstands scrutiny from counterparties, auditors, and regulators alike.

